ISPTEC — INSTITUTO SUPERIOR POLITÉCNICO DE TECNOLOGIAS E CIÊNCIAS

English III — Resolução passo a passo

1.ª Prova Parcelar · 1 de Dezembro de 2025

CURSO	TURMA	DOCENTE	DURAÇÃO
Engenharia de Petróleo	EPT	Anito Pereira	Prova parcelar (5 partes × 2,5 valores)

Regras gramaticais usadas e significado das abreviaturas

Estrutura	Fórmula	Exemplo
Voz passiva	sujeito + **to be** (no tempo do verbo activo) + **particípio passado** (+ *by* + agente)	*Rock samples **are collected** by geologists.*
1.º condicional (situação real/possível no futuro)	**If** + presente simples, **will** + infinitivo	*If he **doesn't wear** PPE, they **won't let** him in.*
2.º condicional (situação imaginária/improvável)	**If** + passado simples, **would** + infinitivo	*He **would increase** your wages if the company **had** the money.*
Verbo modal	modal (*can, may, must, will, would*) + **infinitivo sem *to***	*Students **must study** hard.*
Pergunta com *do/does*	Do/Does + sujeito + infinitivo	*Would you like to go?* (com *would* não se usa *do*)

Significado das abreviaturas e termos

PPE — *Personal Protective Equipment* (equipamento de protecção individual)
ROV — *Remotely Operated Vehicle* (veículo operado remotamente)
kelly — tubo quadrado/hexagonal que roda e transmite o movimento à coluna de perfuração
derrickhand — operário que trabalha na plataforma elevada («torre») da sonda
roughneck — operário de chão da sonda de perfuração
wildcat well — poço exploratório perfurado numa zona ainda não provada
well log — registo do poço (rochas atravessadas e profundidades)
core samples — amostras cilíndricas de rocha extraídas do poço
petroleum trap — armadilha petrolífera (estrutura geológica que retém o petróleo)
past participle (pp) — particípio passado: *conducted, spoken, measured, built*
sb / sth — *somebody / something*

A. Read the text and answer the questions (2,5 marks)

1. The photographs taken from planes or satellites are examined by geophysicists. True or false?

False. The text says that the photographs are examined by **geologists**, not geophysicists.

2. How are exploratory wells sometimes called?

They are sometimes called **«wildcat wells»**.

3. Geochemists analyse samples of rock and soil. True or false? Justify.

False. Geochemists analyse samples of **surface water and soil** (not rock samples) for small amounts of oil and gas that show oil or gas reserves. Rock samples are collected and analysed by **geologists**.

4. What is a well log?

A well log is **a record of the rocks and the depths at which they are found**. A new one is created each time a new well is drilled.

5. What do geologists do to construct a map of the area between the wells?

They **use the information from different well logs** to construct a map of the area between the wells. This process is called **«geological reasoning»**.

B. Match the research methods 1–3 with a–d (2,5 marks)

Research method	Matches with	Why
1. Geological methods	**d) aerial survey**	Geologists use photographs from planes/satellites and collect rock samples
2. Geochemical methods	**c) chemical analysis**	Geochemists analyse water, soil and gas samples (gas chromatograph)
3. Geophysical methods	**a) seismic survey** and **b) magnetic survey**	Geophysicists measure physical properties of the Earth (sound waves, magnetic field)

1 – d · 2 – c · 3 – a, b

C. Correct the sentences (2,5 marks)

N.º	Wrong sentence	Corrected sentence	Rule
1	*Do equipment with porous surfaces can be thoroughly decontaminated?*	**Can equipment with porous surfaces be thoroughly decontaminated?**	A pergunta com modal (*can*) faz-se invertendo o modal; não se usa *do*.
2	*Moving pipes mays trap your hand and crush it.*	**Moving pipes may trap your hand and crush it.**	Os modais não levam *-s* na 3.ª pessoa.
3	*The rotating kelly will to catch your loose clothing if you are too close.*	**The rotating kelly will catch your loose clothing if you are too close.**	Modal + infinitivo **sem** *to*.
4	*Do you would like to go to the theatre next month?*	**Would you like to go to the theatre next month?**	*Would* já faz de auxiliar: não se junta com *do*.
5	*Petroleum engineering students must studying hard.*	**Petroleum engineering students must study hard.**	Modal + infinitivo (não *-ing*).

D. Put these words in the correct order (2,5 marks)

N.º	Words	Correct sentence
1	Read / carefully / the / data / seismic	**Read the seismic data carefully.**
2	Have / I / on the rig / a good post	**Have I a good post on the rig?** (ou: *I have a good post on the rig.*)
3	Job / a / good / and / it / it's / I / like	**It's a good job and I like it.**
4	The / them / is / talking to / roughneck	**The roughneck is talking to them.**
5	Can / you / help / I	**Can I help you?**

Estrutura: frase afirmativa = Sujeito + Verbo + Complemento + Advérbio de modo (*Read the seismic data **carefully***). Pergunta com modal = Modal + Sujeito + Verbo (*Can I help you?*). Em «Job? ... it?» os pontos de interrogação parecem ser artefactos de impressão/fotografia; por isso a resposta 3 é afirmativa.

E. Transform from Active to Passive (2,5 marks)

Fórmula: objecto do activo → sujeito do passivo + **be** (mesmo tempo) + **particípio passado** + (*by* + antigo sujeito).

N.º	Active	Passive
1	Geologists conduct an aerial survey from a plane or satellite.	**An aerial survey is conducted by geologists from a plane or satellite.**
2	They speak English in Namibia.	**English is spoken in Namibia.**
3	A magnetometer measures magnetic fields.	**Magnetic fields are measured by a magnetometer.**
4	They built three schools last year.	**Three schools were built last year.**
5	Geologists on the ground collect rock samples and analyse them.	**Rock samples are collected and analysed by geologists on the ground.**

Em 2 e 4 o agente (*they*) é indefinido/desconhecido e por isso omite-se (*by them* seria desnecessário). Tempos: *conduct, speak, measure, collect* → presente simples → **am/is/are + pp**; *built* → passado simples → **was/were + pp**.

F. Complete the sentences — first conditional (2,5 marks)

Fórmula: If + presente simples, **will / won't** + infinitivo.

N.º	Complete sentence
1	If he **doesn't wear** (not wear) PPE, they **won't let him** (not let him) on the platform.
2	They **will start drilling** (start drilling) in the dry season if the geological results **are** (be) good.
3	If you **use** (use) existing pipelines, you **will have** (have) to increase capacity.
4	If **there is** (there to be) no water flow, you **will be allowed** (be allowed) to use open cut.
5	How **will** you **use** (use) your money if you **win** (win) Elephant bet?

G. Complete the sentences — second conditional (2,5 marks)

Fórmula: If + passado simples, **would / wouldn't** + infinitivo.

N.º	Complete sentence
1	He **would increase** (increase) your wages if the company representative **had** (have) the money.
2	What **would** the derrickhand **do** if the mud **didn't have** (not have) all the specifications?
3	The plant manager **wouldn't be** (not be) happy if the projects **weren't** (not be) in accordance with the regulations.
4	The pilot **would check** (check) the rig structures if the ROV **were** (be) operational.
5	If the well **was/were** (be) already drilled, the field engineer **would collect** (collect) the data.

Atenção: no 2.º condicional, o verbo *to be* usa **were** para todas as pessoas (*if I were, if the ROV were*); *was* é aceite em linguagem informal.

H. Writing — choose a topic (10 or more lines) (2,5 marks)

a) Safety and risk management

Safety and risk management are essential in the oil and gas industry because drilling operations involve high pressure, flammable gases, heavy equipment and many workers. Every company must identify the hazards before the work starts. This process is called risk assessment, and it helps managers to decide how dangerous a task really is.

The first line of protection is personal protective equipment (PPE). Workers must wear helmets, safety glasses, gloves, steel-toe boots and fire-resistant clothing at all times on the rig. Loose clothing is dangerous near rotating equipment such as the kelly, because it can be caught and pull the worker into the machine.

Second, companies must give training to all employees. They should know the emergency procedures, the evacuation routes and the location of the fire-fighting equipment. Regular drills prepare the crew to react quickly if there is a gas leak, a fire or a blowout.

Finally, equipment must be inspected and maintained regularly. Blowout preventers, pipes and pressure gauges are tested, and every incident or near-miss is reported so that the same mistake does not happen again. If companies follow these rules, accidents will be reduced, the environment will be protected and workers will go home safely. In conclusion, safety is not a cost: it is an investment in people, in production and in the future of the industry.

b) Refining process (alternative topic)

Refining is the process that transforms crude oil into useful products such as petrol, diesel, kerosene, lubricants and asphalt. First, the crude oil is heated in a furnace and sent to the distillation column. Inside the column, the different hydrocarbons separate according to their boiling points: light gases rise to the top, while heavy residues stay at the bottom. This is called fractional distillation.

After distillation, other processes improve the quality and the quantity of the products. In cracking, large molecules are broken into smaller and more valuable ones. In reforming, the structure of the molecules is changed to produce high-octane petrol. Finally, impurities such as sulphur are removed by treatment processes, so that the fuels burn more cleanly. If the refinery works efficiently, it will supply the market with safe and clean fuels.